

WINSTON-SALEM REYNOLDS, NC, USA

WMO: 723193

Lat: 36.134N

Lon: 80.222W

Elev: 296

StdP: 97.82

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93807

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-4.9	-18.8	0.7	-3.5	-16.3	0.9	-1.3	9.2	3.2	8.3	3.9	2.7	320	0.389

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	33.4	23.0	32.4	22.8	31.3	22.4	24.7	30.4	24.1	29.5	23.6	28.7	2.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.3	26.6	22.5	17.8	26.2	22.2	17.5	26.0	76.0	30.5	73.6	29.8	71.7	28.8	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.8	5.7	DB	-10.9	35.3	2.8	1.4	-12.9	36.3	-14.5	37.2	-16.1	38.0	-18.2	39.0	(4)
(5)			WB	-11.9	25.9	2.3	1.6	-13.5	27.0	-14.8	28.0	-16.1	28.9	-17.7	30.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	4.3	6.3	10.3	15.6	20.0	24.2	25.7	25.1	21.6	15.7	10.2	6.5	(6)
(7)		DBStd	8.60	5.70	5.12	5.29	4.57	3.75	2.70	2.32	2.40	3.30	4.28	4.67	4.76	(7)
(8)		HDD10.0	580	192	126	63	8	0	0	0	0	0	6	54	132	(8)
(9)		HDD18.3	1888	435	337	253	105	29	1	0	0	11	103	246	368	(9)
(10)		CDD10.0	2586	16	22	72	176	310	425	486	468	347	182	60	22	(10)
(11)		CDD18.3	853	0	1	4	23	80	176	228	209	108	22	2	1	(11)
(12)		CDH23.3	7251	1	2	32	221	633	1549	2096	1808	744	154	11	1	(12)
(13)		CDH26.7	2583	0	0	2	40	168	584	855	681	223	29	1	0	(13)
(14)	Wind	WSAvg	2.4	2.6	2.7	3.0	2.9	2.5	2.2	2.0	1.9	2.2	2.2	2.3	2.3	(14)
(15)	Precipitation	PrecAvg	1167	95	70	109	94	94	111	122	109	113	85	75	90	(15)
(16)		PrecMax	1615	172	134	214	179	210	205	206	200	240	318	186	165	(16)
(17)		PrecMin	831	21	31	21	13	28	22	63	36	15	0	9	31	(17)
(18)		PrecStd	163	40	27	50	45	42	42	38	38	59	70	43	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	22.8	26.4	29.8	32.0	34.4	35.1	34.5	32.8	29.6	25.0	22.2	(19)
(20)			MCWB	15.4	16.0	15.3	17.9	21.1	22.4	23.7	23.4	21.0	21.2	16.8	15.9	(20)
(21)		2%	DB	17.9	19.7	23.9	27.8	30.1	32.8	33.5	32.9	30.9	27.0	22.4	19.0	(21)
(22)			MCWB	13.6	13.5	15.0	17.4	20.3	22.6	23.5	23.3	21.5	19.3	15.4	15.5	(22)
(23)		5%	DB	16.0	17.6	21.6	26.1	28.6	31.5	32.3	31.8	29.2	24.9	20.4	16.8	(23)
(24)			MCWB	12.0	12.2	13.6	16.6	20.1	22.2	23.1	23.0	20.9	17.9	15.2	13.2	(24)
(25)		10%	DB	13.3	15.0	19.1	24.0	27.1	30.1	31.1	30.3	27.6	22.9	18.2	14.3	(25)
(26)			MCWB	9.3	9.5	12.4	15.8	19.2	21.5	22.8	22.5	20.5	17.5	13.7	10.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	17.6	18.2	20.6	22.9	24.8	25.5	25.7	24.1	23.0	19.6	18.3	(27)
(28)			MCDB	19.3	20.7	23.3	26.1	28.4	30.5	32.0	31.1	28.6	25.7	21.2	20.3	(28)
(29)		2%	WB	15.1	15.5	16.6	19.3	21.8	23.9	24.8	24.6	23.3	21.6	17.8	16.1	(29)
(30)			MCDB	17.3	18.6	21.1	25.1	27.3	29.9	30.9	30.0	27.4	25.1	20.2	18.2	(30)
(31)		5%	WB	12.6	13.2	15.3	18.1	21.0	23.2	24.1	24.0	22.6	20.0	16.5	14.0	(31)
(32)			MCDB	15.1	16.3	19.8	23.3	26.2	29.1	29.8	29.0	26.2	23.2	19.5	16.1	(32)
(33)		10%	WB	9.9	10.6	13.8	17.0	20.3	22.6	23.7	23.5	21.8	18.4	14.6	11.4	(33)
(34)			MCDB	13.1	13.9	17.9	22.0	25.2	28.1	29.0	28.3	25.2	21.9	17.6	13.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.6	10.3	10.8	11.5	10.7	10.0	9.4	9.3	9.5	10.7	11.0	9.3	(35)
(36)			MCDBR	11.9	13.2	13.0	13.4	12.2	11.6	10.8	10.9	11.4	12.6	13.1	11.4	(36)
(37)			MCWBR	8.3	8.9	7.0	5.9	4.7	4.1	3.5	3.6	4.2	6.1	7.8	8.0	(37)
(38)		5% WB	MCDBR	10.6	11.5	11.5	11.4	10.1	10.1	9.7	9.2	9.1	9.5	10.3	9.7	(38)
(39)			MCWBR	8.7	9.4	7.1	5.9	4.5	3.9	3.7	3.5	4.0	5.6	8.0	8.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.304	0.318	0.348	0.391	0.433	0.470	0.512	0.504	0.409	0.365	0.320	0.306	(40)	
(41)		taud	2.506	2.462	2.420	2.304	2.213	2.147	2.057	2.057	2.321	2.423	2.508	2.534	(41)	
(42)		Ebn at Noon	897	923	920	892	855	820	782	779	845	856	870	872	(42)	
(43)		Edh at Noon	81	95	109	129	143	153	166	163	119	98	80	75	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.52	3.18	4.24	5.30	5.86	6.30	5.87	5.37	4.50	3.68	2.79	2.15	(44)	
(45)		RadStd	0.17	0.39	0.28	0.42	0.48	0.46	0.38	0.27	0.41	0.46	0.23	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+92

Nomenclature: See separate page